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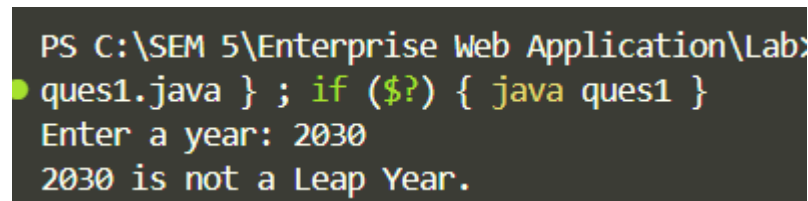
Ques 1: Write Java Program to Check Leap Year

Code:

```
import java.util.Scanner;
public class ques1 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a year: ");
        int year = sc.nextInt();

        if ((year % 4 == 0 && year % 100 != 0) || (year % 400 == 0)) {
            System.out.println(year + " is a Leap Year.");
        } else {
            System.out.println(year + " is not a Leap Year.");
        }
        sc.close();
    }
}
```

Output:



```
PS C:\SEM 5\Enterprise Web Application\Lab>
ques1.java } ; if ($?) { java ques1 }
Enter a year: 2030
2030 is not a Leap Year.
```

Ques 2 : Write Java Program to Check Whether a Character is Alphabet or Not

Code:

```
import java.util.Scanner;
public class ques2 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a character: ");
        char ch = sc.next().charAt(0);

        if ((ch >= 'a' && ch <= 'z') || (ch >= 'A' && ch <= 'Z')) {
            System.out.println(ch + " is an alphabet.");
        } else {
            System.out.println(ch + " is not an alphabet.");
        }
        sc.close();
    }
}
```

Output:

```
● ques2.java } ; if ($?) { java ques2 }  
Enter a character: s  
s is an alphabet.
```

Ques3 : Write Java Program to Find Factorial of a Number

Code :

```
import java.util.Scanner;  
public class ques3 {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
        System.out.print("Enter a number: ");  
        int n = sc.nextInt();  
  
        long fact = 1;  
        for (int i = 1; i <= n; i++) {  
            fact *= i;  
        }  
  
        System.out.println("Factorial of " + n + " = " + fact);  
        sc.close();  
    }  
}
```

Output :

```
($?) { javac ques3.java } ; if  
Enter a number: 5  
Factorial of 5 = 120
```

Ques 4 : Write Java Program to Display Fibonacci Series

Code :

```
import java.util.Scanner;  
public class ques4 {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
        System.out.print("Enter number of terms: ");  
        int n = sc.nextInt();  
        int t1 = 0, t2 = 1;  
        System.out.print("Fibonacci Series: ");  
        for (int i = 1; i <= n; i++) {  
            System.out.print(t1 + " ");  
            int sum = t1 + t2;  
            t1 = t2;  
            t2 = sum;  
        }  
        sc.close();  
    }  
}
```

Output :

```
Enter number of terms: 8
Fibonacci Series: 0 1 1 2 3 5 8 13
```

Ques 5 : Write Java Program to Find GCD of two Numbers

Code :

```
import java.util.Scanner;
public class ques5 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter first number: ");
        int a = sc.nextInt();
        System.out.print("Enter second number: ");
        int b = sc.nextInt();

        int x = a, y = b;
        while (y != 0) {
            int temp = y;
            y = x % y;
            x = temp;
        }

        System.out.println("GCD of " + a + " and " + b + " = " + x);
        sc.close();
    }
}
```

Output :

```
Enter first number: 196
Enter second number: 116
GCD of 196 and 116 = 4
```

Ques 6 : Write Java Program to Find LCM of two Numbers

Code :

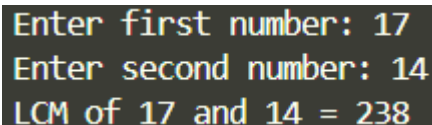
```
import java.util.Scanner;
public class ques6 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter first number: ");
        int a = sc.nextInt();
        System.out.print("Enter second number: ");
        int b = sc.nextInt();

        int gcd = findGCD(a, b);
        int lcm = (a * b) / gcd;

        System.out.println("LCM of " + a + " and " + b + " = " + lcm);
        sc.close();
    }

    static int findGCD(int a, int b) {
        while (b != 0) {
            int temp = b;
            b = a % b;
            a = temp;
        }
        return a;
    }
}
```

Output :



```
Enter first number: 17
Enter second number: 14
LCM of 17 and 14 = 238
```

Ques 7 : Write Java Program to Count Number of Digits in an Integer

Code :

```
import java.util.Scanner;
public class ques7 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter an integer: ");
        int n = sc.nextInt();

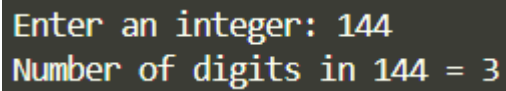
        int num = Math.abs(n);
        int count = 0;

        if (num == 0) {
            count = 1;
        } else {
            while (num != 0) {
                num /= 10;
            }
        }
    }
}
```

```
        count++;
    }
}

System.out.println("Number of digits in " + n + " = " + count);
sc.close();
}
}
```

Output :



```
Enter an integer: 144
Number of digits in 144 = 3
```

Ques 8 : Write Java Program to Reverse a Number

Code :

```
import java.util.Scanner;
public class ques8 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a number: ");
        int n = sc.nextInt();

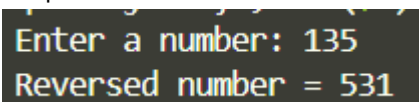
        int num = Math.abs(n);
        int reversed = 0;

        while (num != 0) {
            int digit = num % 10;
            reversed = reversed * 10 + digit;
            num /= 10;
        }

        if (n < 0) {
            reversed = -reversed;
        }

        System.out.println("Reversed number = " + reversed);
        sc.close();
    }
}
```

Output :



```
Enter a number: 135
Reversed number = 531
```

Ques 9 : Write Java Program to Calculate the Power of a Number

Code :

```
import java.util.Scanner;
```

```
public class ques9 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter base: ");
        double base = sc.nextDouble();
        System.out.print("Enter exponent (non-negative integer): ");
        int exp = sc.nextInt();

        double result = 1;
        for (int i = 0; i < exp; i++) {
            result *= base;
        }

        System.out.println(base + " ^ " + exp + " = " + result);
        sc.close();
    }
}
```

Output :

```
Enter base: 6
Enter exponent (non-negative integer): 3
6.0 ^ 3 = 216.0
```

Ques 10 : Write Java Program to Check Palindrome

Code :

```
import java.util.Scanner;
public class ques10 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a number: ");
        int n = sc.nextInt();

        int num = n, reversed = 0;
        while (num != 0) {
            int digit = num % 10;
            reversed = reversed * 10 + digit;
            num /= 10;
        }

        if (n == reversed) {
            System.out.println(n + " is a Palindrome.");
        } else {
            System.out.println(n + " is not a Palindrome.");
        }
        sc.close();
    }
}
```

Output :

```
Enter a number: 14541
14541 is a Palindrome.
```